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#CycloneGaja – Rank based Credibility Analysis System in social media during the crisis

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Abstract

Online Social Networking (OSN) is a platform, which allows users to impart information to other people. Twitter plays an instrumental role in public and political conversations related to the particular crisis events. This research work aims to rank the credibility of the tweets associated with the Natural Disasters and identifies the tweets that spread fake information. The authors propose a Credibility-Analysis-System to rank tweet messages according to their credibility. The authors have compared the performance of tweet credibility with state-of-art of machine learning algorithms such as Naive Bayes classifier, SVM rank algorithm, and Random Forest classifier. The experimentation has been performed on the real-time dataset collected during the Gaja cyclone event. The results present the Random Forest classifier to be the most suitable algorithm for credibility analysis with an accuracy of 87%.

Keywords: Credibility analysis; Machine learning algorithms; Twitter user features;Twitter content features;Cyclone Gaja

1. Introduction

Social media is a powerful tool to share information related to social issues like a natural disaster, election period and breaking news [1]. This media also spread the misinformation to society and it becomes the main threat to our society [2]. Twitter is one of the social networking services, on which dynamic users interact and post short messages called tweets.

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In India, the statistics show that the number of active users on Twitter is growing fast as compared to previous years. Twitter has played a vital role to spread the breaking news during natural disasters such as the Kerala flood. The authors have collected real-time tweets during the Gaja cyclone. In Tamilnadu, on November 11, the deep

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depression strengthened into a cyclonic storm and was given the name *Gaja* by the IMD. The cyclone Gaja generated a tremendous tweet dataset, contributed, by people and institutions reporting the impact of the crisis [5]. A major challenge faced in such a crisis period is the spread of rumor and highly unrelated information [8][9]. The authors have modeled a ranking system that determines the degree of credibility in the tweet text. The term credibility can be understood as "the quality of being trusted and believed in" [7]. A tweet is said to be credible if a user would trust or believe that the information conveyed through it is true [3]. The major contribution of this research work is to check the credibility of the tweets by using a ranking algorithm with 40 tweet features. The authors propose a Credibility-Analysis-System (CAS) model to rank the twitter post based on its credibility. The major contribution of work are

- CAS model to rank the tweet based on its credibility.
- Compare the accuracy of the credibility with different machine learning algorithms.

Section 2 presents the related work about credibility analysis. Section 3 describes tweet collection, data pre-processing and feature value computation. Section 4 discusses and evaluates the results of the proposed system. In section 5 the features are evaluated based on metrics such as accuracy, precision, recall. Section 6 concludes the research findings and presents the future scope.

2. Related Work

Alrubaian et.al. proposed a credibility assessment system to analyze and assess the credibility of the tweet post. This system has four techniques, namely, reputation-based model, feature ranking algorithm, credibility ranking model and user expertise model. The tweet users are automatically ranked using the reputation-based technique. In weighing methodology, the instance features are ranked according to their relative importance. The authors examine the level of extracted features like user-level, tweet-level, and hybrid-level features to analyze the credibility. The tweets are classified as credible or non-credible with the aid of feature ranking algorithm and credibility classification engine [11]. Aditi Gupta et al. developed a semi-supervised ranking algorithm to determine the credit score for six crisis events. The tweets are annotated at two levels, first level tweets have annotated based on information related to the event and the selected 45% of the related tweets further annotated with respect to the credibility. In credibility modeling, they first extract the features from the tweets and apply the semi-supervised ranking model to score the credibility of the content. They have developed a real-time web application to provide the credit score for all tweets. The TweetCred performance is evaluated based on the effectiveness and response time [7]. Zubiaga et al. determine the veracity of the tweet content based on credibility, that perceived from a variety of features from a hurricane event. This system performance was evaluated using an accuracy measure [12]. Gupta et al characterize the spread of misinformation on Twitter with the aid of attributes namely temporal, user and source. They had proposed a linear regression model to predict the rumored content based on user activities. They analyzed Twitter accounts on how many new accounts are created and how many accounts are suspended during hurricane events [13]. Sonu et al. to rank the Facebook post based on credibility. They proposed the A-C-T (Accuracy-Clear-Timely) classification model with 152 features to check the credibility [10].

3. Credibility Analysis System

The proposed rank based Credibility Analysis System has 4 major modules namely tweet collection, pre-processing, data annotation and feature extraction. This system provides the rank for credible tweets based on credit scores for the real-time event #cyclonegaja. The architecture of the proposed system is presented in fig 1.

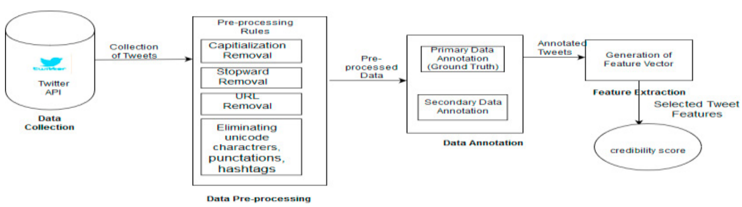

Fig. 1.Architecture for Credibility Analysis System

3.1 Data Collection
The tweets were collected using Tweepy API, which is a streaming API provided by Twitter. This tool collects data in real-time and updates new data once per hour. The authors have collected a large volume of tweets under various hashtags and user IDs namely, #cyclonegaja, @TNSDMA, and #SaveDelta. The tweets were collected over a two week period from 12th Nov 2018 to 25th Nov 2018 using Tweepy API. A total of 60,572 tweets were collected over a two-week period, with tweets from around 26,500 unique users.

3.2 Data Pre-processing
Preprocessing of data is the process of preparing and cleaning the tweets for classification. This process reduces the noise in the Twitter message and helps to filter the credible messages and speed up the classification process. The preprocessing rules are Convert to lowercase, RT removal, Replacement of User-Mentions, URL Replacement, Hash Character Removal, Removal of Punctuations and Symbols, Lemmatization and Stop word Removal.

3.3 Data Annotation
A huge chunk of the process depends heavily on manual annotation of the tweets according to the content presented in a tweet message. This is a very crucial part of the process and the accuracy of the result relies heavily on this. The tweets are annotated in two levels namely primary and secondary. In primary annotation, the relevant information is identified based on the ground truth given by the government officials. The #cyclonegaja dataset was annotated as follows:

R1-Twitter data contains relevant information connected to the event.
R2-The Twitter data is associated with the event but contains no information.
R3-The Twitter data is not associated with the event.

The authors have select only R1 tweets for secondary annotation. In the second phase of annotation, the tweets are ranked according to their credibility of the tweet message shared by the user. Based on the rank the #cyclonegaja event tweets are classified into three categories as follows:

C1-Absolutely credible
C2-Marginally credible
C3-Absolutely incredible

3.4 Feature Extraction
Tweet feature extraction is carried out to identify certain attributes or features which might prove to be useful in performing the credibility analysis of the tweets in consideration. Feature extraction involves reducing the number of features and select data from the subset to improve the accuracy of text classification. It has generated the feature vector based on the tweet features.

3.5 Tweet Credibility
The tweet's credibility is analyzed based on the ranking system [4]. The data annotation of tweets is very important for checking the credibility. These annotation results are used to rank the twitter messages from 1-5. The ranking is done by mapping the annotation labels to integer values, the following transformation was used:

5=Relavant and absolutely credible (class R1.C1)
4= Relevant and marginally credible (R1.C2)
3=Relevant and absolutely incredible (R1.C3)
2=Associated (R2)
1=Not associated (R3)

The above ranking metric shows that the quality of the tweet's credibility based on the score. This score shows better results to analyze the credibility of the tweets. Tweet analyzers may adopt this ranking system to identify tweets during crisis time in order to identify the fake news.

4. Results and Discussions

4.1 Data Collection

The credibility has been analyzed upon the hashtags #cyclonegaja, @TNSDMA, and #SaveDelta. The following fig 2 illustrates the tweets are collected for the above hashtags during the time interval of 12/11/2018 to

25/11/2018. A total of 60,572 tweets were collected over a two-week period, with tweets from around 26,500 unique users.

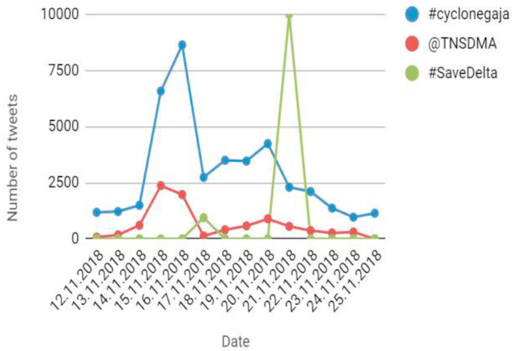

Fig. 2. Tweet Collection during Gaja cyclone

It can be observed from the combined graph that most of the tweets were posted during the time period of 14.11.2018 - 17.11.2018, with the exception of the tweets containing '#SaveDelta', which peaked between 20.11.2018 - 22.11.2018. The following fig 3 shows the geographical locations of the users who posted about the event in a given period of time. This graph was obtained by feeding in the inputs of the locations of various tweets into the Google Maps API, which was fetched along with various other attributes during the tweet collection process.

Fig. 3. Geo-located Tweets

4.2 Feature Extraction

Feature extraction refers to the process of deriving certain attributes from the contents of the tweets. For example, length, number of words, etc. Tweet contains two features namely tweet features and user features. The authors extract 40 tweet features and then applied to the CAS system to check and rank the credibility of the tweet.

4.3 Pre-processing

The collected tweets are further pre-processed based on the rules discussed in the previous section and Table 1 presents the rule-wise as well as the overall pre-processed sample tweet captured based on #cyclonegaja.

TABLE I. PRE-PROCESSING OF SAMPLE TWEET

Sample Tweet	RT @tnsdma: Cyclone alert for north Tamilnadu #CycloneGaja IMD Bulletin No.18 https://t.co/hU7LvrdDJ2	
Rule No.	**Rule Name**	**Pre-Processed Tweet**

1	Convert to lowercase	rt @tnsdma: cyclone alert for north tamilnadu #cyclonegaja imd bulletin no.18 https://t.co/hu7lvrddj2
2	RT removal	@tnsdma: cyclone alert for north tamilnadu #cyclonegaja imd bulletin no.18 https://t.co/hu7lvrddj2
3	Replacement of User Mentions	user: cyclone alert for north tamilnadu #cyclonegaja imd bulletin no.18 https://t.co/hu7lvrddj2
4	URL Replacement	user: cyclone alert for north tamilnadu #cyclonegaja imd bulletin no.18 url
5	Hash Character Removal	user: cyclone alert for north tamilnadu cyclonegaja imd bulletin no.18 url
6	Removal of Punctuations and Symbols	user cyclone alert for north tamilnadu cyclonegaja imd bulletin url
7	Lemmatization	user cyclone alert for north tamilnadu cyclonegaja imd bulletin url
8	Stopword Removal	user cyclone alert for north tamilnadu cyclonegaja imd bulletin url
9	Unwanted Whitespaces Removal	user cyclone alert for north tamilnadu cyclonegaja imd bulletin url
Pre-processed Tweet	User cyclone alert for north tamilnadu cyclonegaja imd bulletin url	

4.4 Credibility Analysis System

This section deals with the tweet credibility analysis system and to investigate which ML algorithm is best suited for credibility analysis. Support Vector Machine (SVM) rank works by mapping tweets to a high-dimensional feature space so that data points can be categorized, even when the data are not linearly separable [7]. In machine learning, Naive Bayes (NB) classifiers are a family of simple "probabilistic classifiers" based on applying Bayes' theorem with strong (naive) independence assumptions between the features [5]. The ideas behind the Random Forest (RF) algorithm are to generate multiple little trees from random subsets of data. After creating n number of trees in a random manner, we will be able to obtain more cluttered decision boundaries than the simple lines [6]. This algorithm incorporates a metric, referred to as NDCG which is used to arrive at better conclusions regarding the classification results. When the results of the three classifying algorithms are compared, the random forest classifier offers a better accuracy rate compared to the SVM rank algorithm and Naive Bayes classifier. The state-of-art classification algorithms SVM, NB, and RF along with the nDCG values have been applied. In order to rank the credibility of the tweets, the authors use the metrics of Discounted Cumulative Gain(DCG) and normalizedDCG (nDCG). The DCG measures the highly relevant text related to the event. It measures logically down weights the gain received from the Twitter data and examine the long ranked result list by adjusting the discounting value. To normalize the ranking system the authors sort all the relevant documents in the dataset based on the relative relevance and produce highly possible DCG through the position p known as Ideal DCG(IDCG). To rank the twitter data , the authors first computed the DCG value . The DCG is defined as,

$$DCG = \sum_{i=1}^{|n|} \frac{rel(x_i)}{\log_2(i+1)} \quad (1)$$

where n is a number of elements and rel() is referred to as some relevance function of the ith element in a given list.

For comparison of multiple features, a normalizedDCG is derived by

$$nDCG = {}^{DCG}/_{IDCG} \quad (2)$$

The performance of the credibility analysis is normalized relative to the ideal, based on the graded relevance judgments. It was very useful to evaluate data with multi-valued classification. Comparison results for three algorithms with NDCG values are shown in fig 4.

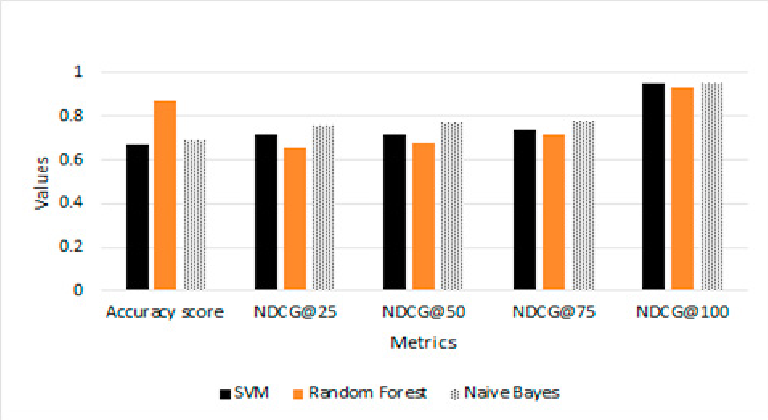

Fig. 4. Comparison of the credibility analysis system with nDCG Values @25, @50, @75, @100

5.Evaluation Metrics

The performance of the credibility analysis system is computed with the aid of metrics accuracy, precision, recall, f1-score. The training and testing ratio for each algorithm is 75% and 25%.The accuracy of this model is defined as the number of tweets are correctly classified as credible tweets. Precision is defined as the ratio of the total number of retrieved documents that are actually relevant. The recall is the fraction of relevant tweets associated with the retrieved over the total amount of relevant tweets. The tweet features are evaluated using machine learning algorithms. The authors have analyzed the metrics based on the outcome classes from the data annotation section. The evaluation metrics for precision, recall, and F1-score are shown in fig 5, 6 and 7 respectively.

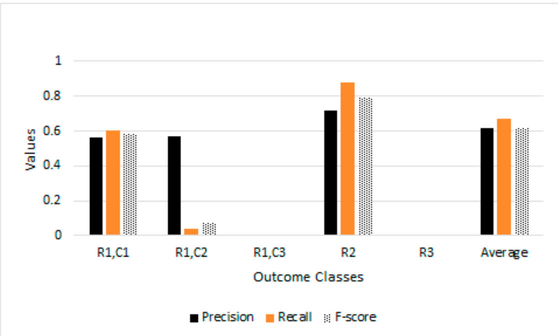

Fig. 5. Evaluation metrics – Support Vector Machine

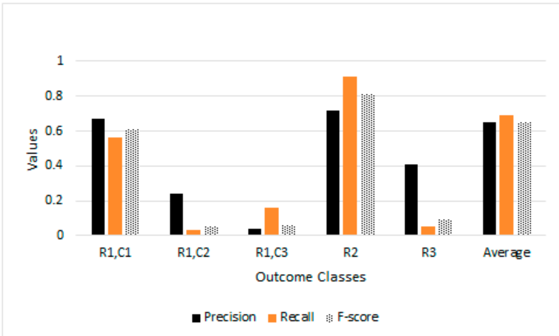

Fig. 6. Evaluation metrics – Naïve Bayes

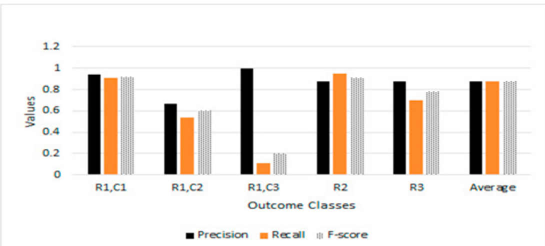

Fig. 7. Evaluation metrics – Random Forest

These metrics are used to identify the best credibility ranking system. Precision, recall, and F1-Score are high for random forest algorithms compare to the other two algorithms. It is because the random forest algorithm performs well with the mixture of numerical and categorical features to improve the credibility rank system. As a result, tweet features contribute much to the ranking system. This algorithm performs well among others as they have the highest evaluation metrics. From the result, it is concluded that tweet features with the random forest algorithm perform well for scoring credibility.

6.Conclusions

During natural disaster time, Twitter is one of the important media for sharing information like government announcements, requests for rescue. But some of the Twitter users spread the rumors knowing or unknowingly. It is important to check the credibility of the tweet messages. The authors present a credibility analysis system to detect trustworthy information during crisis time. This system has provides a rank to the credibility of the tweet. It mainly relies heavily on the manual annotation of the data and this plays a huge role in deciding the accuracy and precision of the model that is proposed.The authors have focused on analyzing the credibility of real-world tweets into a particular category by using machine learning classifiers like NB, SVM, and RF algorithms. The performance result shows that the RF algorithm computes credibility scores better than NB and SVM.

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